

NAKKA SIVA PRASAD

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Career Objective

To obtain a full-time position as an **Embedded Engineer** where I can apply my expertise in **Embedded C programming, STM32 development, peripheral interfacing and communication protocols** to build efficient, real-time microcontroller-based systems.

Education

Rajiv Gandhi University of Knowledge and Technologies, Srikakulam

- B.Tech in ECE | 2020 – 2024 | CGPA: 8.5
- Intermediate (MPC) | 2018 – 2021 | CGPA: 8.9

Z.P.H.S

- SSC | 2019 | GPA: 10.0
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Professional Experience

Embedded Engineer (Full-Time)

Laxven Systems, Hyderabad | Aug 2025 – Present

Project: DPWCS (Distributed Power Wireless Control System)

- Developed RGB display software (**BT817Q**) with multiple UI screens.
- Wrote display-to-CPU communication code as per data packet format.
- Integrated display with system modules and ensured reliable data exchange.
- Worked on STM32 firmware with UART and SPI communication.

Project: KAVACH (TCAS – Train Collision Avoidance System)

- Designed hardware and developed software for the RCC (Radio Communication Card) subsystem, which communicates with the main motherboard.
- Worked on Radio Unit consisting of Interface Card and Power Supply Card.
- Power Supply Card included 110V to 24V DC-DC converter, 24V to DC-DC converter, and power monitoring circuits.
- Interface Card included RS485 to RS232 converter sections for radio modem communication.

Project: UHF Radio (Ongoing)

- Leading end-to-end development of a UHF Radio system as a single owner.
 - Designed system with Control Card (STM32) and RF Card architecture.
 - Developing firmware for UART/SPI communication, control logic, and RF interfacing.
 - Handling schematic design, integration, testing, and validation
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Junior Embedded Engineer Intern
Laxven Systems, Hyderabad | Aug 2024 – Aug 2025

Project: KAVACH – Train Collision Avoidance System (TCAS)

Hardware and Software Testing Loco KAVACH Testjig

- Studied schematics and datasheets to understand system design.
- Developed firmware, wrote code, and tested hardware modules.

Technical Skills

Programming Languages: Embedded C, C, Assembly (8051 – Basics)

Microcontrollers: STM32, ESP32, Arduino, 8051

Operating Systems / RTOS: Basics of FreeRTOS

Tools & IDEs: STM32CubeIDE, Arduino IDE, Code::Blocks, MATLAB, Tinkercad

Communication Protocols: UART, SPI, I2C, CAN, RS-232, RS-485

Hardware Tools: Oscilloscope, Logic Analyzer, Multimeter, Function Generator, dB Meter

PCB Design: KiCad v7, Proteus.

Debugging Tools: ST-Link, Docklight, Serial Monitor.

Core Competencies

Schematic analysis | Datasheet interpretation | Hardware bring-up | Peripheral validation |
Protocol testing & debugging | Firmware debugging | Technical documentation

Projects

STM32-Based Projects

- **Automatic Gate System Using STM32:** IR sensor triggers servo-controlled gate using interrupts.
- **Password-Based Security System Using STM32:** Correct password opens gate, incorrect triggers buzzer alert; implemented keypad, debouncing, verification logic, and hardware feedback.
- **Event Logger Using STM32 with RTC:** Lights automatically ON/OFF at preset times; RTC ensures accurate real-time events.

ESP32-Based Projects

- **CAN Communication Between Two ESP32 Boards:** Bi-directional sensor data transfer and OLED display via CAN protocol.

Arduino-Based Projects

- **Power Generation via Speed Breakers:** Generated electricity using piezoelectric sensors from road traffic.
 - **Smart Dustbin:** Automated lid operation using ultrasonic sensor and servo motor.
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Certifications & Training

- Embedded C – Udemy
- Embedded Systems – Ignited Mind Organization
- 8051 & 8085 Microprocessors – Bharath Acharya Education

Personal Traits

Team Leader | Creative | Flexible | Quick Learner | Dedicated | Integrity | Hardworking